

Communication is a mediated effect

Separating signals from exploited cues in dual-use action channels, with an application to active electrosensing

Abstract

An action that both senses and broadcasts is hard to classify. A weakly electric fish’s discharge illuminates its own surroundings and simultaneously announces its presence to anyone with the receptors to detect it, and the same structure arises for active sonar, a robot’s laser sweep, or any observable act of sensing. Whether such an act is *communication* or a cue that others merely exploit is routinely settled with three diagnostics—positive signalling, causal influence, and the payoff cost of ablating the channel—and we show that each is individually foolable, on dyads whose communication status is known by construction. Positive signalling is 0.934 bits for a sender that nobody listens to, and causal influence (31.4 nats) and payoff-ablation (-75.5) both fire for a channel that carries no information at all, because deleting a message takes the receiver outside its own training distribution. We therefore define communication as a *natural indirect effect*: the sender’s private state acting on the receiver’s payoff through the message, estimated by transmitting the message the sender would have produced in an independently drawn world while the true world is held fixed. We prove it is zero whenever the message is independent of the sender’s state or the receiver ignores it, and it is the only one of the four measures that answers all four validation dyads correctly (-95.5 where communication exists, -0.27 and -0.90 where it does not). Applied to the electric organ discharge, the picture inverts the received one. Silencing a fish costs it -0.40 food items per episode, of which -0.40 is its own lost electrolocation and 0.11 its lost detectability by others, a split that holds between 83% and 135% across group sizes, arena scales, episode lengths and network widths, and that reappears in a minimal dual-use benchmark containing no fish physics at all. The channel carries decodable information about food ($\Delta R^2 = 0.152$) and moves receiver policies $13\times$ the noise floor, yet its mediated effect is indistinguishable from zero. It is an exploited cue, not a signal.

1 Introduction

A weakly electric fish navigates by emitting brief electric pulses and reading the distortions that nearby objects impose on the resulting field [1, 2]. The same pulses propagate far beyond the range at which they are useful for electrolocation, and other fish—equipped with dedicated knollenorgan electroreceptors tuned to conspecific discharges—can detect them at up to a metre [3]. An EOD is therefore simultaneously a private sensory probe and a public event. Which of these functions explains why the fish emits when it does is the central question of electric-fish neuroethology, and it is an instance of a much older problem in animal communication: distinguishing a *signal*, whose production has been shaped by its effect on receivers, from a *cue*, which receivers merely exploit [4, 5].

Singh et al. [6] introduced a computational framework in which agents with biophysically grounded electrosensory and electromotor systems are trained by multi-agent reinforcement learning to forage collectively. Their agents reproduce several features of real fish, and interventions on the model—ablating electroreceptor classes, silencing an agent’s EOD, and altering the food distribution—show that discharges influence group foraging and social spacing. They define communication operationally, as “EOD signals that are sensed by conspecifics via Knollenorgans and that influence their behavior.”

That definition is a statement about effects, and effects are cheap. The emergent-communication literature has repeatedly shown that behavioural association between one agent’s output and another’s behaviour is not sufficient evidence of signalling: a receiver may be responding to something correlated with the message, and a sender’s output may be informative without having been shaped by that informativeness at all [7, 8]. Lowe et al. [7] formalise the two requirements as *positive signalling* (the message carries information about the

sender’s state) and *positive listening* (the message causally changes the receiver’s behaviour), and note that either can occur without the other.

Our setting makes the problem sharper than a generic emergent-communication study, because the channel is not a dedicated cheap-talk symbol but a physical act with an unavoidable private benefit. A single discharge has three consequences: **reafference**, the electric image of nearby objects cast onto the emitter’s own mormyromasts, which is why the behaviour exists at all; **illumination**, the same field polarising objects near a *neighbour*, whose mormyromasts can read those images for free—collective electrosensing, demonstrated in real fish by Pedraja and Sawtell [9], and the paradigm case of a cue; and **detection**, the pulse registering on the neighbour’s knollenorgans, which encode emitter identity and bearing and are the only one of the three that could be a signal. Silencing a fish removes all three at once, so any behavioural consequence is uninterpretable—the muted fish may forage worse because it can no longer see, its neighbours because they lost free illumination, or because a signal they attended to has gone. This is not an artefact of the model; it is why EOD-ablation experiments in real fish are hard to read.

Contributions. We make the three consequences independently switchable and ask what changes. (i) A **channel algebra** for dual-use signals: reafference, illumination and detection become three masks on one motor act, letting us build pulses that are private, exploitable-but-undetectable, detectable-but-uninformative, phantom, replayed from another context, or scrambled in time or sender identity. (ii) An **interventional** causal influence: because we own the simulator, the counterfactual of Jaques et al. [8] is evaluated with the do operator directly—re-rendering the receiver’s input under $\text{do}(e_j=1)$ and $\text{do}(e_j=0)$ with every other cause on its factual path—rather than approximated with a model of other agents. (iii) A **cue/signal discriminator**: we train matched populations whose only difference is whether the focal agent can be *heard*, holding its own reception fixed, and measure the divergence of their emission policies against seed-to-seed divergence within each world. (iv) An **economics**: metabolic cost per discharge, predators that localise prey by discharges, cooperative versus competitive harvests, tunable social range, persistent identity. (v) An **assay validation**: dyads that are communicating by construction, against which every metric is checked before it is trusted, plus a **decoupled channel** and a **non-fish dual-use benchmark** to separate what is about electric fish from what is about dual use.

A note on wording. Our populations adapt within a training run by policy-gradient learning under a fixed reward; we do not run an evolutionary outer loop over heritable traits. We therefore write *emerge under learning* throughout, and reserve claims about evolutionary history for the Discussion, where we say explicitly what our design can and cannot license.

2 Related work

Measuring emergent communication. Most work equips agents with a dedicated, costless channel that has no other use, so that any use of it is communicative by construction and the open question is what code appears [10–13]. Within that setting, Lowe et al. [7] showed that message–behaviour association is insufficient evidence and introduced positive signalling and positive listening; Jaques et al. [8] proposed causal influence, estimated through a learned model of other agents; Eccles et al. [14] added training biases that make communication more likely to appear; Bogin et al. [15], Chaabouni et al. [16] study compositionality and context-independence. Our contribution to this line is not another metric but an audit: we show that these diagnostics are individually foolable, exhibit a case where positive signalling is *higher* in a population whose receivers are deaf and where causal influence *rises* as the channel becomes less informative, and validate the whole battery against dyads of known communication status before applying it.

Dual-use channels, and the cue/signal distinction. Our setting inverts the usual one: the channel is a physical act with a large private benefit, so the question is whether *any* of its use is communicative—the situation for active sonar, a robot’s laser sweep, or any action that both senses and is observable, and closer to the biological norm, since signals are generally thought to be ritualised from behaviours that had other functions. Noukhovitch et al. [17] show that influence under competition can reflect manipulation or cue-reading, the same identification problem we attack with the payoff and production tests. The distinction

we operationalise is Scott-Phillips’s: a signal’s production has been shaped by its effect on receivers, a cue’s has not [4, 5]. Costly-signalling theory [18–20] explains how honesty is maintained in a signal that exists; our cost sweep asks whether cost brings one into being, and finds it does not.

Electric fish. Singh et al. [6] built the reference simulator and training setup we reimplement, reporting that discharges shape group foraging and spacing; our departure is definitional and causal, since their definition is effect-based and their one intervention removes three consequences at once. The biology we add is established: collective electrosensing [9], corollary-discharge cancellation [21], discharge energetics [22, 23], and electroreceptive eavesdroppers [24, 25].

3 Communication as a natural indirect effect

The three diagnostics in common use fail on the dyads of §4.1, and they fail for a shared structural reason: none of them asks the question that defines communication. Positive signalling looks only at the sender, and is satisfied by a sender talking to nobody. Causal influence and positive listening look only at the receiver, and are satisfied by a receiver attending to noise. Ablation asks what happens when the message is removed, which is a different question from what the message conveys—and removal takes the receiver outside the support of its own training distribution, which is precisely why it fires for a channel that carries nothing.

What communication requires is that the receiver’s payoff depend on the sender’s private state *through* the message. That is a mediation question, and it has a standard causal answer [26, 27].

Definition. Let S be the sender’s private state, M the message it emits, and u_R the receiver’s return. S influences u_R along two kinds of path: direct ones, since the sender also moves, occupies space and consumes resources, and the mediated path through M . Write $M(s)$ for the message the sender’s policy would emit in state s , and $u_R(s, m)$ for the receiver’s return when the world is in state s and the transmitted message is m . The *natural indirect effect* of the message is

$$\text{NIE} = \mathbb{E}_{s, s' \sim p(S)} [u_R(s, M(s')) - u_R(s, M(s))], \quad (1)$$

with s' drawn independently of s . Operationally: hold the world at s , so every direct pathway is untouched, and transmit the message the sender *would have* produced in an independently drawn world. We estimate it by running matched pairs of episodes that share initial conditions and the sensory-noise stream, and differencing within pair.

Two properties do the work.

Proposition 1 (Insensitivity to uninformative channels). *If $M \perp S$, then $\text{NIE} = 0$.*

Proof. If the message does not depend on the sender’s state then $M(s')$ and $M(s)$ are identically distributed for every s , so the two terms of Eq. 1 have equal expectation. $\square$ $\square$

Proposition 2 (Insensitivity to unattended channels). *If the receiver’s policy does not depend on M , then $\text{NIE} = 0$.*

Proof. Then $u_R(s, m) = u_R(s)$ for all m , and the difference vanishes pointwise. $\square$ $\square$

Proposition 3 (Deletion is not mediation). *The ablation contrast $\mathbb{E}[u_R(s, \emptyset)] - \mathbb{E}[u_R(s, M(s))]$, where $\emptyset$ denotes a removed or zeroed message, can be non-zero even when $M \perp S$.*

Proof. By counterexample, realised in §4.1: a receiver whose policy conditions on M commits to a target on the basis of a message that carries no information about S . Removing M leaves it with an input outside the support of M , so it does not commit at all, and its return falls by -75.5. Yet $M \perp S$, so by Proposition 1 the mediated effect is zero and no information has been lost. $\square$ $\square$

Propositions 1 and 2 say the NIE is zero in exactly the two ways a system can fail to communicate while satisfying a standard diagnostic, and Proposition 3 says why the payoff test that looks closest to ours is not equivalent to it. The distinction is that $M(s')$ is drawn from the message’s own marginal: the receiver always sees a message it could have seen, and only the correspondence between message and world is broken.

Estimation. Eq. 1 needs $M(s')$, the message the sender’s policy would have produced in a counterfactual world. In a simulator this is directly available. In the referential game we resample the referent, recompute the sender’s message from it, and transmit that while the true referent and the food stay where they were. In the naturalistic foraging world the sender’s state is not a single latent variable, so we use the trajectory-level analogue: the message actually transmitted is replaced by the pulse train the same policy produced in an independently drawn arena. Marginal rate, burst structure and inter-pulse-interval statistics are preserved exactly—these are the same trains that policy emits—while the correspondence between the train and the receiver’s world is destroyed.

What it does not do. The NIE is a statement about payoff, so it will report zero for a channel that is informative and attended but whose use happens not to matter to either party in the environment tested; it measures the *realised value* of the correspondence, not its existence. It also inherits the usual mediation caveat: the decomposition into direct and indirect paths is well defined only if the message can be intervened on without disturbing the direct paths, which is exactly what the channel algebra of §K.2 buys us and which is not available in an unmodified simulator. Finally, like any causal-effect estimate it is bounded by statistical power, so a null must be read against a measured detection floor—which is what the validation dyads provide.

4 Results

4.1 Do the assays work? A dyad that is communicating by construction

Every negative result below rests on an assay reporting “no communication here”. An assay that reported that unconditionally would produce the same paper, so before applying them we check that they fire on a case where communication is not in question.

We embed a Lewis signalling game in the same physics. One fish is immobile and privately observes which of two sites holds food; the other must travel to a site. The sender’s discharges are forced onto a fixed schedule, so pulse timing carries nothing, and the only free channel is a one-bit discharge subtype. Both fish are scored on the receiver reaching the correct site, and the harvest is shared. Crossing an informative sender (subtype = the private cue) against an uninformative one (a bit that is constant within a trial but uncorrelated with the truth), and a receiver that attends to the subtype against one that ignores it, gives four dyads whose communication status we know (Appendix D). Only the informative-and-attended dyad succeeds: 4.00 correct arrivals out of four per episode, against 1.98 for chance.

Each assay behaves as it should, and—more usefully—each one alone is foolable. Positive signalling is 0.934 bits for the informative sender and 0.000 for the uninformative one, but it is just as high when nobody is listening, because it measures the sender in isolation. Causal influence and positive listening separate the attending receiver from the deaf one (31.4 versus 0.000 nats) but are just as high when the channel carries nothing, because a receiver can organise its behaviour around a variable that is pure noise. Even the payoff test is foolable on its own: deleting the subtype costs the attending receiver -173.4 when the subtype is informative and still -75.5 when it is noise, because the receiver has come to depend on it either way.

The mediated effect of §3 answers all four correctly, and it is the only one of the four measures that does (Fig. 1). It is -95.5 (95% CI [-102.9, -87.9]) for the communicating dyad and indistinguishable from zero for the other three: -0.90 when the message is informative but ignored, and -0.27 when it is attended but carries nothing. The contrast with the ablation test on the same dyad is the point of Proposition 3: deleting the message costs that receiver -75.5, because it has organised its behaviour around a variable it can no longer see, while resampling the message from its own marginal costs it nothing, because there was never any correspondence to break.

This also fixes the detection floor. Any null we report later must be read against -95.5—the size of the effect these instruments demonstrably resolve in this simulator.

4.2 What the optimiser can and cannot do

A negative result about communication invites the objection that the learner, not the environment, is the limiting factor. We measured this rather than argued about it, because the answer bounds what the rest of

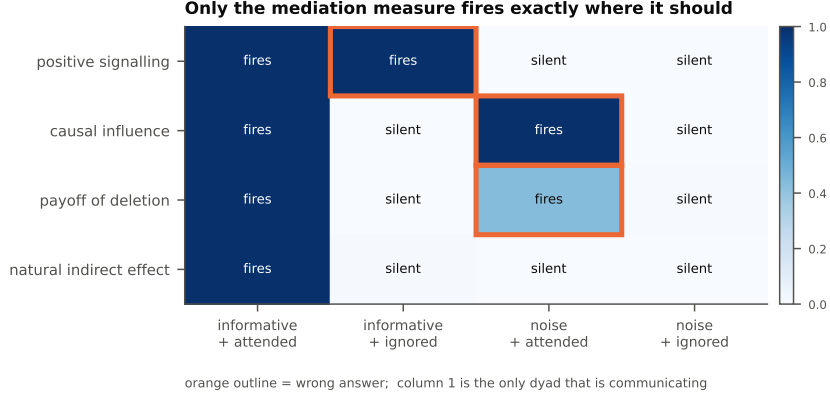


Figure 1: Which diagnostic fires where. Cells are normalised within each row; orange outlines mark answers that disagree with ground truth. Only the first column is communicating. The three standard measures make 3 errors between them; the mediated effect makes none.

the paper may claim.

In the same referential game we trained recurrent MAPPO under three conditions, twelve seeds each at 50×10^6 agent-steps: both sides learning; the receiver learning against a *scripted honest* sender, so the message is already perfectly informative and only reading it must be discovered; and both sides learning with the positive-signalling bias of Eccles et al. [14], which exists to break exactly this deadlock. *None of the thirty-six runs solved the game*: all sit at chance, against 4.00/4 for the hand-coded dyad reading the same observation vector. Reducing the steering noise by an order of magnitude and halving the credit horizon do not move it either (Appendix E).

So shared-parameter recurrent MAPPO—the standard configuration, and the one the reference model uses—does not discover that a single perfectly informative input bit should select between two known targets in an embodied task, even when told what to say. Our results therefore divide in two. The muting decomposition, the foolability of each diagnostic, and the metrics’ behaviour under intervention are measurements on trained policies and do not depend on this. The claims that metabolic cost fails to create a signal, and that a decoupled channel does not either, are claims about what learning arrives at and inherit the bound: these pressures did not produce signalling under a standard learner, not that they could not under a better one.

4.3 A validated, fast reimplementation

The environment is a GPU reimplementation of the reference simulator, checked stage by stage against the original NumPy code in double precision on random scenes. All 11 checks agree to floating-point round-off (maximum relative deviation 6×10^{-10} ; the knollenorgan and ampullary pathways agree bit-for-bit), and it sustains 596k agent-steps per second in aggregate on two GPUs, which is what makes 354 independent training runs affordable. Appendix A gives the per-stage table and records an incidental bug in the reference code: its configuration file specifies a wall-reflection coefficient of 0.95, but the scene builder never forwards it, so the published results in fact use unattenuated images.

4.4 The muting effect is almost entirely private sensing

Agents that can discharge eat 5.12 food items per episode against 1.78 for agents muted from the start of training ($2.9\times$, Hedges’ $g = 4.5$, $p = 0.0001$): the probe does real work, which is the precondition for everything else. Removing the three consequences one at a time during training orders them—no detection 4.71, no illumination 4.31, private probe only 3.92, no reafference 4.27—but from-scratch comparisons confound the value of a channel with the difficulty of learning without it (Appendix B), so we turn to interventions on frozen policies.

The central experiment. We take a trained, intact population and, at evaluation time only, silence one fish—the intervention used upstream—then repeat with each of its three consequences removed separately. Every condition is run from the same episode seed, so arena k has the same initial food layout and the same sensory-noise stream in all of them, and differences are taken within arena.

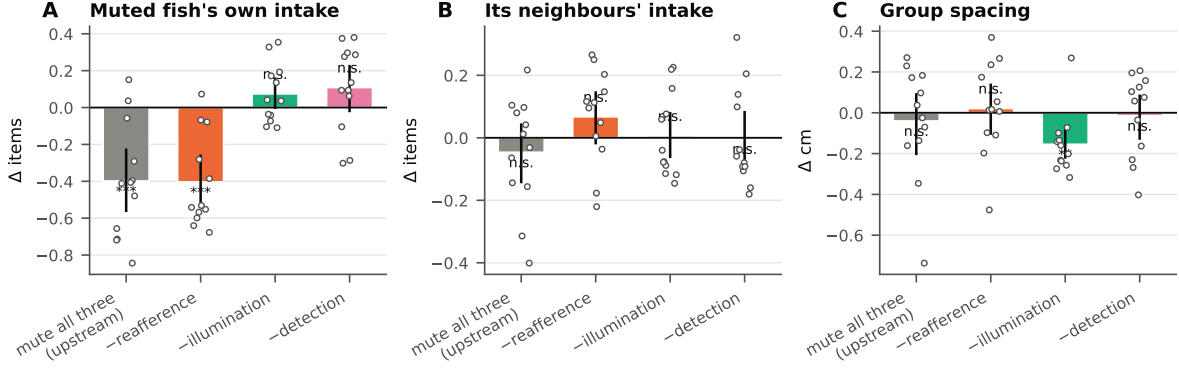


Figure 2: Silencing one fish removes three things at once; separating them reassigns the effect. Within-arena paired differences from the intact channel, frozen policies. Significance is a permutation test over seeds.

Muting one fish costs that fish -0.40 food items per episode (Fig. 2A). Decomposed: removing only its own reafference costs -0.40, removing only its illumination of neighbours costs 0.07, and removing only its detectability costs 0.11. The private share accounts for essentially the whole effect.

The effect on its *neighbours* is smaller still (Fig. 2B): -0.05 items in total, with the illumination share 0.01 ($p = 0.8733$) and the detection share -0.00 ($p = 0.9414$). Group spacing (Fig. 2C) is likewise unchanged within resolution.

This is the paper’s first substantive result, and it is a negative one. When a fish is silenced in this model, what happens is overwhelmingly that the fish has been blinded, not that a conversation has been interrupted. An experiment that reports only the joint effect—as the reference study does, and as EOD-ablation experiments in real fish must—will attribute to communication an effect that is almost entirely private electrolocation.

4.5 The channel nonetheless carries information and moves policies

It would be wrong to conclude that the public channel is inert. It is not.

Discharge probability depends on the emitter’s situation: 0.587 when food is within 10 cm versus 0.457 when it is not ($p = 0.0001$; Fig. 9C). That state-dependence is inherited by the public channel, and a receiver can read it: decoding *only* from the number of pulses a receiver heard from a given emitter in a trailing window, food proximity is recovered with $\Delta R^2 = 0.152$ above a time-shifted baseline ($p = 0.0000$; Fig. 9B). Positive signalling about food, measured as mutual information conditioned on distance-to-neighbour decile so that geometry cannot manufacture it, is 0.085 of the state entropy (Appendix G).

Positive listening holds too. The interventional causal influence of Eq. 7—re-rendering a receiver’s sensory input under $\text{do}(e_j=1)$ versus $\text{do}(e_j=0)$ with everything else held on its factual path—is 0.0026 nats, against a noise floor of 0.00020 nats obtained by re-rendering the *same* intervention under an independent noise draw (Appendix G). The channel moves receiver policies $13\times$ more than sensory noise does.

Both of these diagnostics, however, behave badly here in ways worth recording. Positive signalling is *higher* in populations whose receivers are deaf (0.152) than in populations whose receivers can hear (0.085): a fish modulates its probe rate with its own situation whether or not anybody is listening, so the measure is picking up the sensory schedule, not signalling. This is exactly the artifact Lowe et al. [7] warn of, and it is straightforward to demonstrate once a deaf population is available as a control. Causal influence misbehaves in the other direction: it *rises* 12-fold, from 0.0026 to 0.0303 nats, as metabolic cost is increased—that is, precisely as the pulse train becomes rarer and less informative (Appendix G). A rare discharge is more surprising, so switching it on or off moves the receiver’s policy more, even though the train as a whole now tells the receiver less. Influence per message and information per unit time are different quantities, and in this system they are anti-correlated.

4.6 ... but its mediated effect is zero

Positive signalling and positive listening are both satisfied, so on the effect-based definition this is communication. The mediated effect says otherwise.

Transmitting the pulse train the same policy produced in an independently drawn arena—identical marginal rate, burst structure and inter-pulse statistics, no correspondence with the receiver’s actual world—changes the receivers’ return by 0.21 (95% CI $[-0.98, 1.50]$), which is 0.49% of what they earn, and the sender’s by 1.88. Swapping in trains recorded in the opposite food context gives the same answer. Against the -95.5 the same measure returns on the validation dyad, that is a null roughly 449 times smaller than the effect these instruments resolve.

The other channel manipulations agree. Temporal scrambling, identity scrambling and phantom pulses injected at rates from 0.05 to 0.6 all leave returns unmoved (Appendix G). We report these as no effect detectable at this sample size rather than as demonstrated equivalence: no comparison clears the margin in either direction — the intervals are simply wider than the ± 2.18 margin (cross-arena replay, $+0.21$ (90% CI $[-2.87, +3.31]$); temporal scrambling, $+0.63$ (90% CI $[-2.43, +3.68]$); removing the public channel entirely, -0.70 (90% CI $[-3.77, +2.35]$)). We therefore report these as *no effect detectable at this sample size*, and explicitly not as demonstrated equivalence. The honest statement is that the design rules out payoff effects larger than roughly a tenth of the intact return, and is not powered to rule out smaller ones.

A receiver can read the emitter’s situation off its pulse train, and does adjust its policy when the train changes, but the correspondence between train and world is worth nothing to either party. On the definition of §3 the discharge is an exploited cue.

4.7 Cost and predation regulate the probe without creating a signal

Adding a metabolic price gives a clean dose-response: discharge probability falls from 0.495 at $\lambda = 0$ to 0.032 at $\lambda = 0.08$. Eavesdropping predation pushes it lower again at every cost and mortality falls with it, so crypsis works; cooperative harvesting sustains a higher rate than competition at matched cost. We expected economy to make the train informative, since a budget must be spent where it pays. The opposite happens: information about food falls monotonically from $\Delta R^2 = 0.098$ to 0.010, and it keeps falling after normalising per pulse (0.127 to 0.087). The learned response to a price is a lower baseline rate, not a smarter schedule. Silence behaves the same way: discharge probability is essentially unchanged by a predator within 25 cm (-0.004 , 95% CI $[-0.007, +0.000]$) and danger is near-undecodable from the public channel ($+0.010$ AUC), so agents manage average exposure rather than timing gaps. Appendix C gives the full grid and the regime labels.

4.8 Sender shaping, with reception held fixed

Our first attempt at the production-side test compared populations trained with and without knollenorgans, and gave SSI = 0.419 at zero cost. That contrast is confounded, and in a way worth spelling out because it is easy to make. Disabling knollenorgans removes *reception*. It therefore asks whether a fish behaves differently when it can hear others, not whether it behaves differently when others can hear *it*. The direction of the effect gives the confound away: hearing populations discharge *less* than deaf ones (0.547 versus 0.611), which is free-riding—a fish that can hear its neighbours’ probes cuts its own costly sampling—and not audience design, which would push emission up or restructure it.

The yoked condition holds reception fixed and cuts audibility instead. Receivers get knollen input with matched statistics, drawn from another arena, so no sender’s pulses reach anybody while every receiver still has a realistic social input stream. Measured this way, sender shaping is 0.045 at zero cost (Appendix G), and discharge rate is 0.576 live versus 0.583 yoked. Emission policy is therefore no more different between the live and yoked worlds than between seeds of the same world. Holding reception fixed, being listened to leaves no detectable trace on how the discharge is scheduled, which is what it means for the pulse to be a cue rather than a signal.

4.9 Freeing the channel is not sufficient either

A negative result is only as good as its positive control, and to this point every manipulation left the discharge welded to its sensory function. We therefore gave the agents a second variable: a one-bit discharge subtype that rides on the pulse, so sending still costs a pulse and still exposes the fish to predators, but whose value changes nothing about the emitter’s own electric image, nothing about how the pulse illuminates neighbours, and nothing about detectability. Only the content is decoupled, not the act. Agents mark 52% of their discharges with the subtype, which is indistinguishable from the 50% an unmodulated bit would give, and it carries 0.0141 bits about food proximity above its permutation null. Deleting the subtype leaves the pulse, and therefore every sensory consequence, completely intact, so any payoff change is attributable to content alone. Across the four ecological settings: standard $\times$ compet., +0.51 (95% CI $[-0.45, +1.40]$, $p = 0.292$); standard $\times$ cooper., +0.54 (95% CI $[-1.81, +2.99]$, $p = 0.640$); sparse $\times$ compet., -0.76 (95% CI $[-1.76, +0.20]$, $p = 0.165$); sparse $\times$ cooper., +0.51 (95% CI $[-0.88, +1.95]$, $p = 0.512$). No cell survives correction for multiple comparisons. Freeing the content of a discharge from its sensory function was not, on its own, sufficient to make that content payoff-relevant under any pressure we applied.

4.10 The same decomposition in a world with no electric fish in it

If the result is about dual-use action channels rather than about electroreception, it should reappear wherever an action senses and broadcasts at once. We built a minimal environment with no fish physics in it: agents forage in a unit square, and a PING action has exactly the three consequences a discharge has—it shows the pinger the items near itself, it shows nearby others the items near *them*, and it is registered by everyone as an event with a bearing. A weak always-on passive sense means silencing an agent is not the same as blinding it. The same masks, the same trainer and the same intervention battery apply unchanged.

The decomposition replicates. Agents that can ping collect 13.07 items per episode against 6.03 for permanently silent ones; muting one agent costs it -9.89 items, of which -9.89 (100%) is its own lost refference and -0.06 its lost detectability (Appendix I). Pricing the ping suppresses it from 0.753 to 0.001. Neither the sign nor the rough magnitude of the decomposition depends on the electric-fish biophysics; what it depends on is that one action does two jobs.

4.11 How far the decomposition generalises

The decomposition is not an artefact of one configuration. Repeating it with two and six fish, a 100 cm arena, 1024-step episodes and a 256-unit recurrent policy, the private share of the muting effect stays between 83% and 135% (Table 2), and the detection share remains indistinguishable from zero in every case.

5 Discussion

An effect on a receiver is not a signal. The operational definition used by Singh et al. [6]—an EOD that conspecifics detect and that changes their behaviour—is satisfied in our model at every metabolic cost we tested, including zero. It is satisfied because the knollenorgan channel really does carry information and receivers really do respond to it. Yet the same channel has no detectable consequence for anyone’s payoff: replaying, scrambling or fabricating it moves neither party’s return by an amount we can resolve. We are careful about what that licenses. Our equivalence tests do not clear the 5% margin we set in advance, because the confidence intervals are wider than the margin; the defensible statement is *no effect detectable at this sample size*, not demonstrated equivalence. What makes the negative interpretable is not the width of those intervals but the assay validation of §4.1: the same tests, applied to a dyad that is communicating by construction, return effects one to two orders of magnitude larger than anything we see here.

So the strongest defensible form of the claim is this. A system can satisfy the standard positive-signalling and positive-listening diagnostics without any detectable payoff relevance and without evidence that production was shaped by being heard.

Establishing the production-side half of the argument turned out to need more care than we first gave it. Disabling knollenorgans removes *reception*, so a hearing/deaf contrast asks whether a fish behaves differently when it can hear others—not whether it behaves differently when others can hear *it*. Those are different

questions, and the first is contaminated by free-riding: a fish that can hear its neighbours’ probes cuts its own costly sampling, which is what we in fact observe. The control that isolates being listened to has to hold reception fixed and cut audibility, which is what our yoked condition does.

The muting confound. Silencing a fish is the natural intervention and it is the one available in real experiments, but it removes three things simultaneously. Our decomposition shows that the three shares are not merely different in size—they act on different individuals and in different directions. Losing refference hurts the muted fish; losing the illumination it provided hurts its neighbours; losing its detectability changes how the group is spaced. A single number attributed to “communication” averages over all three. The same caution applies to real ablation work: an EOD-silenced *Gnathonemus* is not a fish with its telephone taken away, it is a fish that has been blinded and has stopped lighting the room for everyone else.

Cost does not manufacture semantics. It is natural to expect that pricing a pulse would turn a probe schedule into an informative one: a budget has to be spent where it pays, and that allocation should make the train diagnostic of the emitter’s state. That is not what happens. Across the whole cost range the pulse train becomes rarer and *less* informative about every target we decoded, and it stays less informative after normalising per pulse. The learned response to a price is a lower baseline rate, not a smarter schedule. Whatever costly signalling does in the classical models [18–20], it presupposes a signal whose honesty is at stake; it does not by itself bring one into being.

What we should not conclude is that a binary pulse leaves no room for a code. A binary event repeated in continuous time supports a large space of temporal codes—inter-pulse intervals, bursts, pauses, event-locked patterns—and mormyrid communication is known to use pulse timing and waveform identity [3, 28, 29]. Our agents *failed to exploit* the temporal code available to them under the pressures we applied. That is a claim about what learning found, not about what the channel permits.

Silence tracks the regime, not the moment. Cessation of discharge is documented in mormyrids as both predator avoidance and submissive display [24, 30], and we expected a momentary version: quiet when the predator is near. It does not appear. Discharge probability is essentially unchanged by a predator within 25 cm (-0.004 , 95% CI $[-0.007, +0.000]$) and danger is near-undecodable from the public channel ($+0.010$ AUC). Agents lower their *baseline* rate wherever predators exist, and mortality falls with it, so crypsis works without timing. The likely reason is sensory asymmetry: our predator hears discharges at 60 cm but is passively detectable only at 25 cm.

Signalling as a public good and a public hazard. Two externalities operate in opposite directions, neither in the reward function: a discharge illuminates neighbours’ worlds, and it also jams their passive sense and draws an eavesdropping predator toward the group. Cooperative harvesting sustains a higher rate than competition at matched cost, consistent with the public-good component being internalised when the harvest is shared. We do not state that as a mechanism: our interventions found no detectable payoff consequence of the public channel, so we cannot also claim neighbours benefit enough from a pulse to explain the rate difference, which may instead reflect altered encounter distributions or an effect below our resolution.

What the discharge lacks is not a channel. The decoupled control tested whether dual use is what blocks signalling: give the fish a variable it can move without degrading its own sensing, and a social function should appear. It does not. Setting that beside the referential assay says what is missing. The assay differs from the foraging world in three ways and only one is the channel; it also gives the sender information the receiver cannot obtain itself, and pays both when the receiver acts on it. In the foraging world a fish finds food with its own probe, so a neighbour’s private state is worth little — there is a channel available and nothing to say over it. Channel factorisation is plausibly necessary for a signal to become distinct from a probe; it is clearly not sufficient. The prediction for biology is that electric communication should be elaborated exactly where one fish reliably knows something a conspecific cannot discover first-hand — dominance, reproductive state, identity — and not where both can simply sense the world.

Relation to emergent-communication work. Most emergent-communication studies equip agents with a dedicated, costless, otherwise-useless channel [10–12], where any use of the channel is communicative by construction and the question is what code appears. Electric fish invert this: the channel is a physical act with a large private benefit, and the question is whether *any* of its use is communicative. We think this is the more common situation in biology, and it is the setting in which a mediation test earns its keep, because it is the setting in which informativeness and influence come for free.

6 Limitations

Six bound the claims above and are set out in Appendix J. The three that matter: our populations adapt by policy-gradient learning within a training run, not by an evolutionary outer loop, so the Sender Shaping Index is a within-lifetime analogue of Scott-Phillips’s criterion rather than the criterion itself; our knollenorgan channel is all-or-none, as upstream, so graded amplitude signalling is structurally impossible in the coupled condition; and the mediated effect is a statement about realised payoff, so it reports zero for a correspondence that exists but happens not to pay in the environment tested.

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Table 1: Agreement between the GPU reimplementation and the reference NumPy implementation, in double precision on random scenes. Error is the maximum absolute deviation normalised by the largest magnitude present.

Stage	Max. relative deviation
Coulomb field of point charges	2.3×10^{-15}
Field of point dipoles	1.4×10^{-14}
Field including first-order wall images	1.9×10^{-15}
Dipoles induced on conductors	1.2×10^{-15}
Charge-limiting of fish body moments	1.2×10^{-15}
Corollary-discharge baseline	1.6×10^{-14}
Ampullary intrinsic baseline	1.3×10^{-14}
Mormyromast, self-image mode	6.0×10^{-10}
Mormyromast, collective-sensing mode	4.9×10^{-11}
Knollenorgan (conspecific detection)	exact (0)
Ampullary (passive DC)	exact (0)

Appendix

A Numerical validation against the reference implementation

Every stage of the sensory pipeline is checked against the original NumPy code in double precision on random scenes (`scripts/validate_physics.py`). Agreement is at floating-point round-off, and the knollenorgan and ampullary pathways agree bit-for-bit.

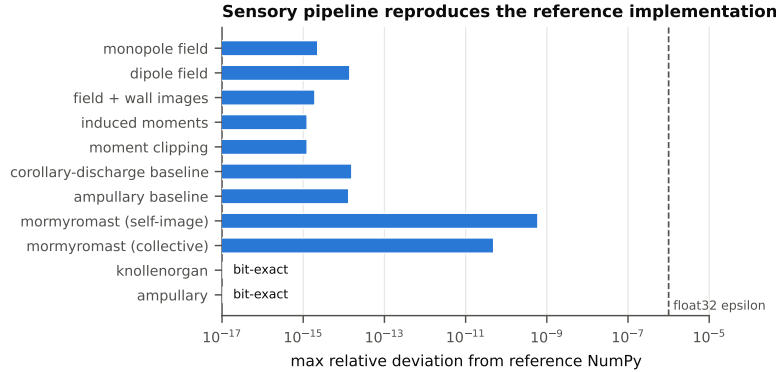


Figure 3: Maximum relative deviation from the reference NumPy implementation over random scenes, in double precision. Bars at the left edge are exactly zero.

An incidental bug in the reference code. Its configuration file sets a wall-reflection coefficient of 0.95, but `ElectricScene.build_reflections` never forwards it to `reflect_sources`, whose default is 1.0. The published results therefore use unattenuated first-order images. We match the code rather than the configuration file; using 0.95 shifts the corollary-discharge baseline by $\sim 1.4 \times 10^{-6}$ relative and changes ampullary readings qualitatively, because those are a small difference of large numbers.

B Training the three channel conditions from scratch

Training is bimodal: a seed either discovers patch foraging or stays near the floor set by passive sensing alone. A seed counts as converged if it reaches half the ceiling of its own task family, a criterion applied identically to every condition within a family; the fraction clearing it is reported as its own outcome, because conditions that remove sensory channels are not only worse on average but less reliably learnable.

C Regime classification

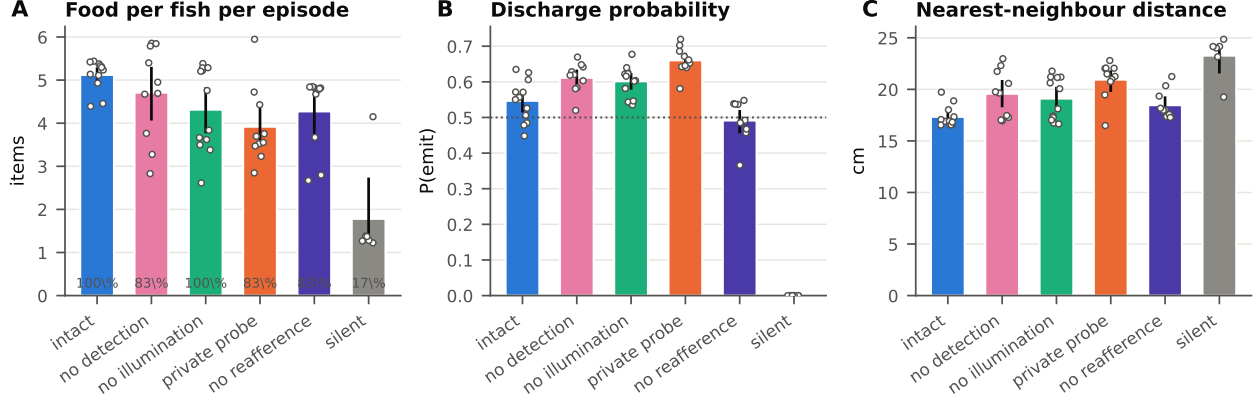


Figure 4: Removing each consequence of a discharge, trained from scratch. Bars are means over independent seeds with bootstrap 95% CIs; points are seeds. Percentages in (A) give the fraction of seeds reaching the learning criterion.

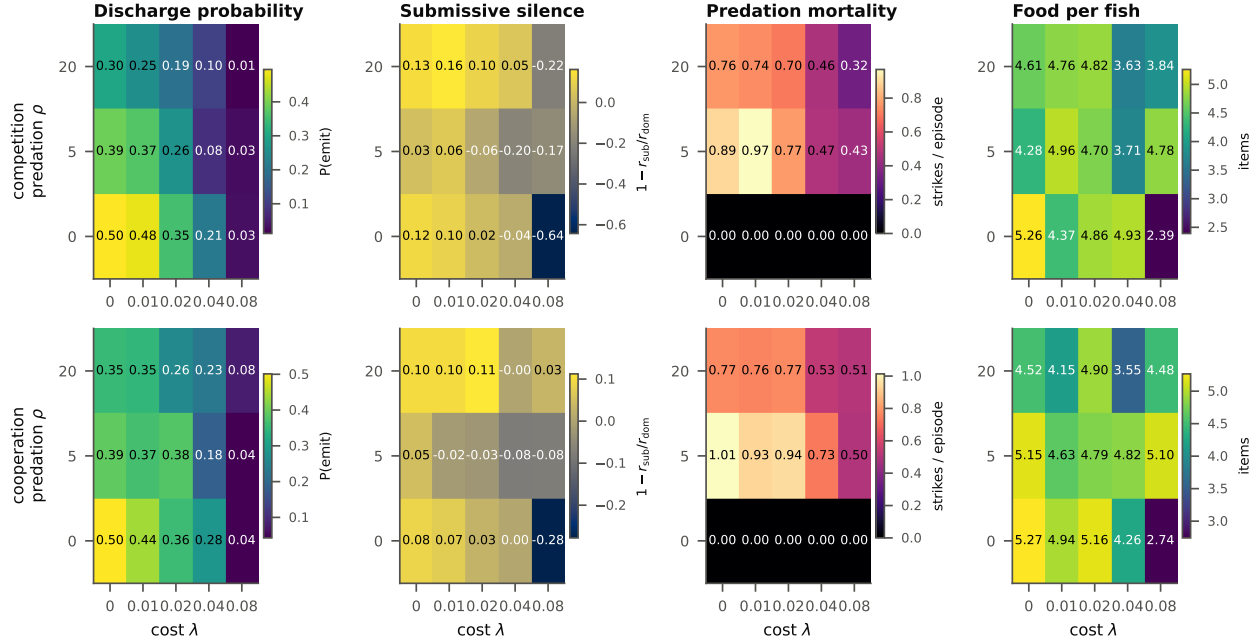


Figure 5: Metabolic cost and eavesdropping predation jointly set discharge policy. Rows are competitive and cooperative harvests; each cell is a mean over independent seeds.

D The validation dyads in detail

E Why the learned referential game fails

The hand-coded receiver reads the subtype and the two site bearings out of the observation vector and reaches the correct site on every trial, so the information is present and a competent policy exists inside the representable class. To check that the learned failure is not an artefact of exploration noise or of the delay between committing to a direction and arriving, we re-ran the receiver-only condition (scripted honest sender) with the initial policy standard deviation on the steering action reduced from $e^{-0.5}$ to e^{-2} , and with the two sites moved from 60 cm apart at a range of 49 cm to 40 cm apart at a range of 25 cm, which roughly halves the credit-assignment horizon and doubles the number of trials per episode. Neither change moves performance off chance:

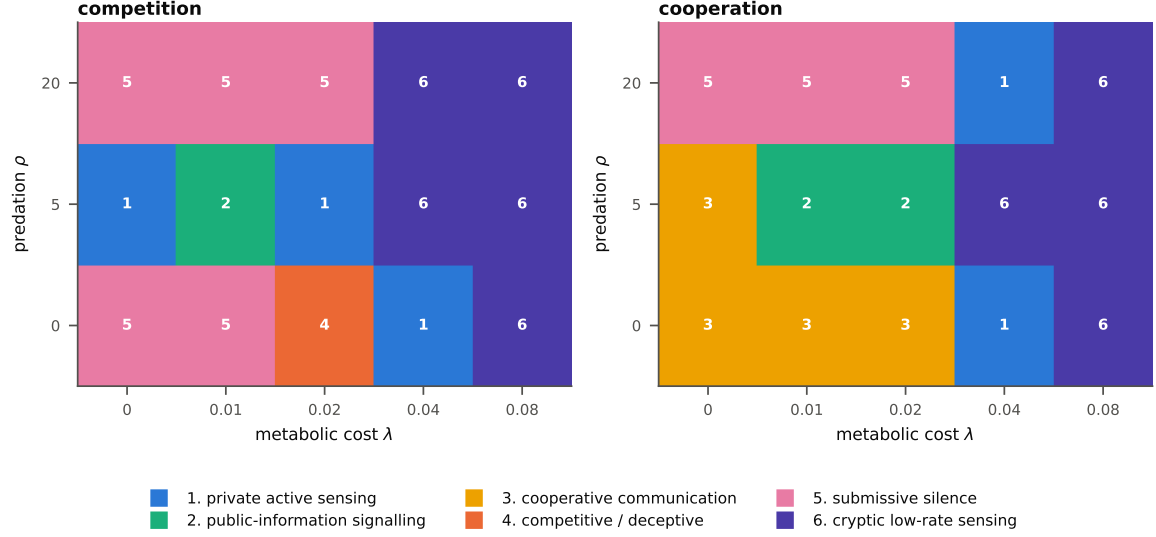


Figure 6: Rule-based regime labels from the pooled metric vector. Thresholds are percentiles of each metric’s own distribution across the grid rather than hand-set values, so the boundaries describe a continuum and are not claims about sharp transitions.

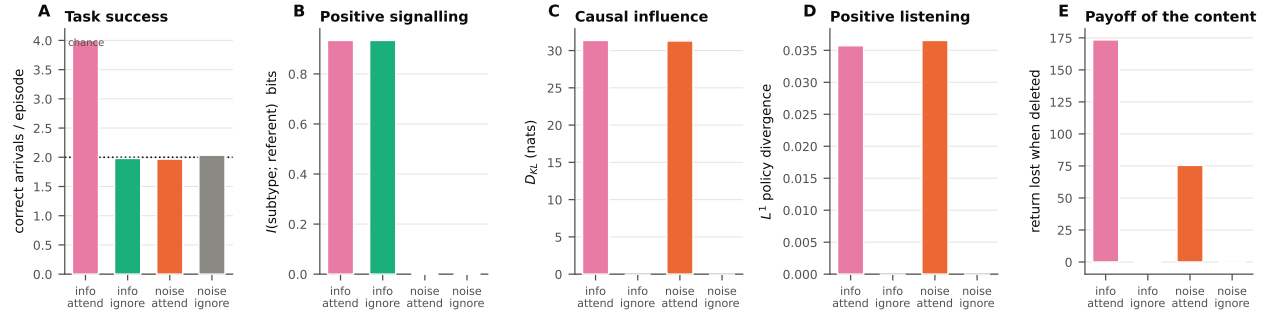


Figure 7: Four dyads in the referential game. Only *info+attend* is communicating; the other three each satisfy some subset of the diagnostics.

Configuration	arrivals / episode	chance
far sites, low steering noise	2.01 of 4	2.0
near sites, default noise	3.00 of 6	3.0
near sites, low steering noise	3.07 of 6	3.0

We therefore attribute the failure to the difficulty of discovering a multiplicative interaction between a single binary input and a continuous steering policy under policy-gradient updates, rather than to the task geometry or the exploration schedule.

F Generalisation of the decomposition

G Message content and sender shaping

H The decoupled signalling variable

I The dual-use benchmark

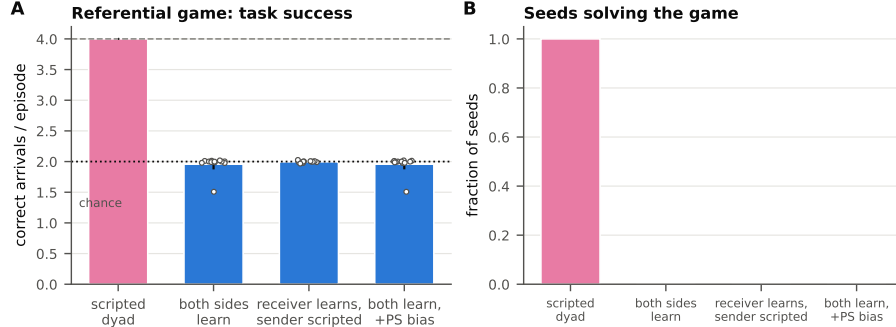


Figure 8: A hand-coded dyad solves the referential game from the same observations; twelve seeds each of three learned configurations, including one with a published bias designed for this problem, do not leave chance.

Table 2: The muting decomposition across scales and architectures. “Private share” is the fraction of the total muting effect attributable to the emitter’s own lost refference.

Condition	seeds	total	–reafference	–detection	private share
baseline (4 fish, 60 cm, 512 steps)	12	-0.40	-0.40	+0.11	101%
2 fish	6	-1.66	-1.58	-0.02	95%
6 fish	6	-0.43	-0.59	-0.08	135%
100 cm arena	6	-0.31	-0.34	-0.02	107%
1024-step episodes	6	-1.10	-0.92	+0.08	83%
256-unit GRU	6	-0.61	-0.57	+0.11	93%

J Limitations in full

We list these plainly because several of them bound the claims above.

1. **Learning is not evolution.** Our populations adapt within a training run by policy-gradient learning under a fixed reward. The Scott-Phillips criterion refers to evolutionary history, and SSI is a within-lifetime analogue: it asks whether the emission policy that learning arrives at depends on whether receivers respond. A genuine evolutionary outer loop over heritable emission traits would test the criterion directly, and we did not run one. We write “emerge under learning” throughout rather than “evolve”.
2. **One arena scale, one group size.** Everything is measured with four fish in a 60×60 cm arena, in which the knollenorgan channel at unit gain reaches every conspecific. The range sweep partially addresses this, but group size is not varied, and the upstream model randomises arena size in a way we deliberately do not.
3. **Pulses are binary and fixed-amplitude.** Mormyrid electric communication uses inter-pulse-interval patterning and species- and individual-specific waveform structure [28, 29]; in wave-type gymnotiforms, amplitude modulations carry honest size information [31]. Our knollenorgan channel is all-or-none, as upstream, so waveform and amplitude signalling are structurally impossible in the coupled condition. This is precisely the limitation the decoupled condition is designed to lift, and it lifts only the crudest version of it: one bit of subtype, not a graded waveform.
4. **Episodes are short.** 512 steps is about 6 s of fish time. Dominance relationships in real fish are established over much longer horizons, so the dominance-related results should be read as within-encounter, not as stable social structure.
5. **The metabolic cost is a reward term, not a physiological model.** We express cost in reward units and report the fraction of the achieved return it consumes so that it can be compared with measured energy budgets, but we do not model the electrocyte physiology that sets the true cost, nor its dependence on amplitude.
6. **The predator is a caricature.** It is guided by discharges alone. Real electroreceptive predators also use vision, mechanoreception and olfaction, so real silence buys less safety than it does here, and the cryptic regime is correspondingly easier to reach in our model than in a river.

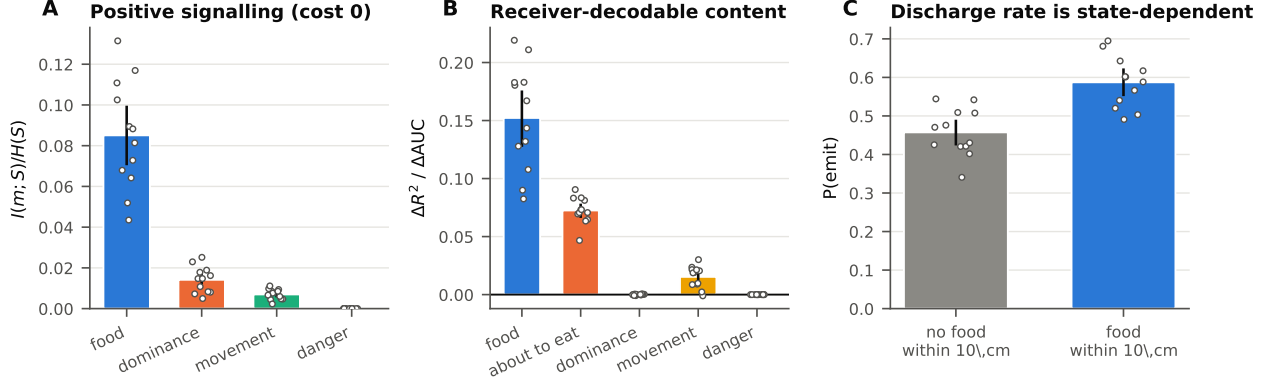


Figure 9: What the pulse train carries. (A) Positive signalling, conditioned on distance-to-neighbour decile. (B) Cross-validated decoding from receiver-observable pulse counts only, minus the time-shifted baseline. (C) Discharge probability by food proximity.

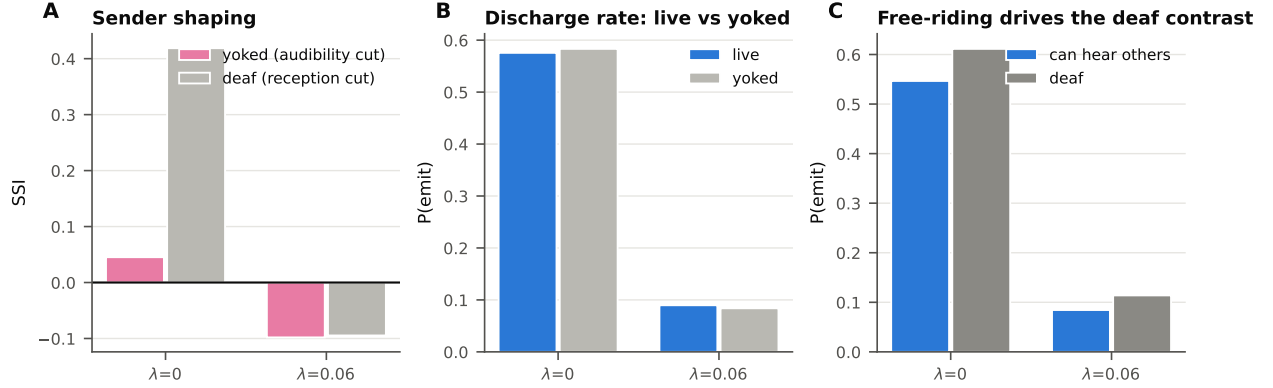


Figure 10: Sender shaping measured with reception held fixed. (A) SSI from the yoked contrast, which cuts audibility only, beside the confounded deaf contrast, which cuts reception. (B) Discharge rate, live versus yoked. (C) The free-riding effect that drives the deaf contrast.

7. Regime labels are a classification imposed on a continuum. The phase diagram is generated by thresholding continuous metrics at percentiles of their own pooled distribution. The boundaries are therefore descriptive, not sharp transitions, and we make no claim about criticality or bifurcation.

K Methods

K.1 Simulator

The environment is a GPU reimplement of the weakly-electric-fish simulator of Singh et al. [6], rewritten so that thousands of independent arenas are advanced as a single set of tensor operations. All electrostatic constants, receptor counts, receptor placements, dynamic ranges, transduction functions, locomotor limits and reward coefficients are transcribed from that codebase; the differences are enumerated in §K.6.

State and kinematics. Each arena holds $F = 4$ fish, $N = 48$ food items and optionally one predator, in a $60 \text{ cm} \times 60 \text{ cm}$ rectangular tank with insulating walls. Fish i has position $\mathbf{x}_i$, heading θ_i and a body size $s_i \in [0, 1]$ that is fixed within an episode and sets a dominance rank. The timestep is $\Delta t = 1/83 \text{ s} = 12.05 \text{ ms}$, matched by the original authors to the social echo-response latency of *Gnathonemus petersii* [32]; episodes last 512 steps ($\approx 6.2 \text{ s}$). Motion is first-order in the commanded velocities,

$$\omega_i = u_i^{\text{turn}} \omega_{\text{max}} (1 + s_i), \quad \mathbf{x}_i \leftarrow \mathbf{x}_i + \hat{u}(\theta_i) \max(u_i^{\text{fwd}}, 0) v_{\text{max}} (1 + s_i), \quad (2)$$

with $v_{\text{max}} = 0.422 \text{ cm per step}$ and $\omega_{\text{max}} = 0.0434 \text{ rad per step}$, taken from the empirical percentiles used upstream. A fish whose proposed position would overlap another fish does not move and receives a collision

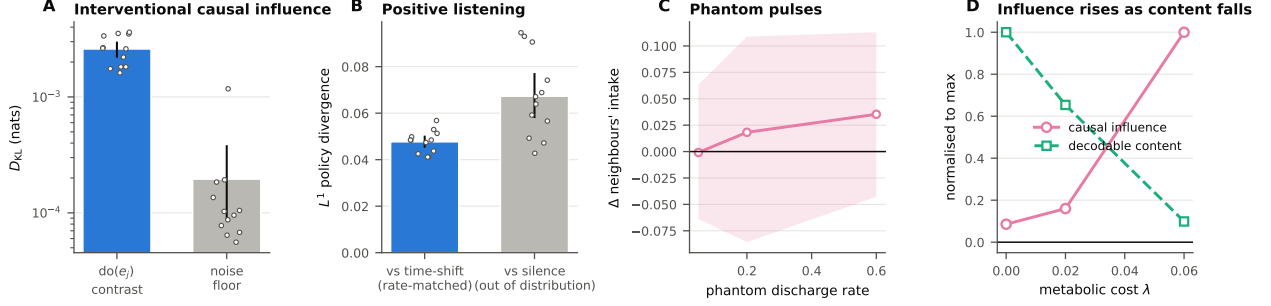


Figure 11: Positive listening, and why it is not enough. (A) Interventional causal influence against its noise floor (log scale). (B) Positive listening against a rate-matched time-shift null and against silence, which is out of distribution. (C) Phantom-pulse dose-response. (D) As metabolic cost rises, causal influence per discharge increases while decodable content falls.

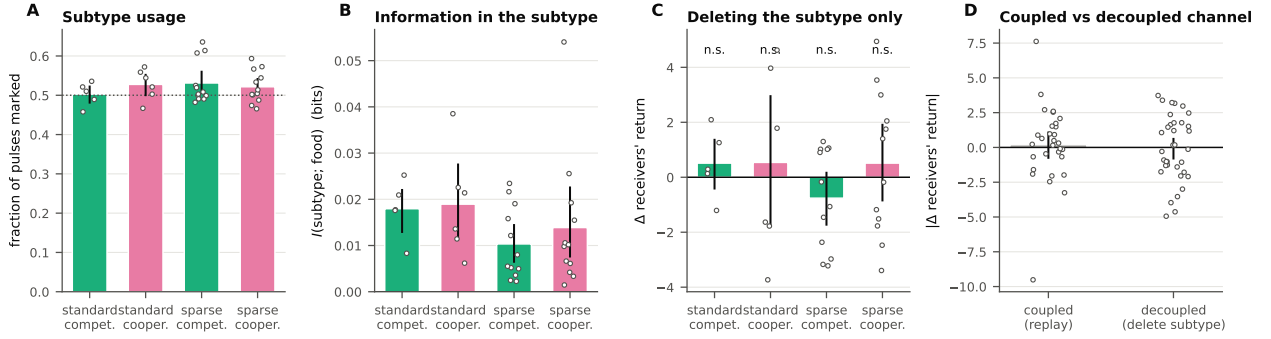


Figure 12: A discharge subtype decoupled from sensing, crossed with competitive and cooperative harvests and with a sparse-patch task in which finding food is the bottleneck. (C) deletes the subtype while leaving the pulse, and therefore all sensing, completely intact — the manipulation the coupled channel does not admit.

penalty; walls clamp position without penalty.

Electric fields. All fields are exact Coulomb superposition, with the distance softening $\varepsilon_m = 10^{-5}$ m applied *before* exponentiation exactly as upstream. A discharging fish carries two monopoles at ± 0.5 cm along its body axis with charges $\pm q_0$, $q_0 = 1.11 \times 10^{-15}$ C [33]:

$$\mathbf{E}_{\text{mono}}(\mathbf{r}) = k \sum_p \frac{q_p(\mathbf{r} - \mathbf{r}_p)}{(\|\mathbf{r} - \mathbf{r}_p\| + \varepsilon_m)^3}, \quad \mathbf{E}_{\text{dip}}(\mathbf{r}) = k \sum_d \left[\frac{3(\mathbf{p}_d \cdot \mathbf{u}_d)\mathbf{u}_d}{\|\mathbf{u}_d\|^5} - \frac{\mathbf{p}_d}{\|\mathbf{u}_d\|^3} \right], \quad (3)$$

with $k = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$ and $\mathbf{u}_d = \mathbf{r} - \mathbf{r}_d$. Fish and food carry permanent intrinsic dipoles (1.11×10^{-23} and 1.11×10^{-24} C m). Food items and non-discharging fish are polarisable insulating spheres,

$$\mathbf{p}_n^{\text{ind}} = 3\varepsilon_0 V_n \chi \mathbf{E}(\mathbf{y}_n), \quad V_n = \frac{4}{3}\pi a_n^3, \quad \chi = -0.5, \quad (4)$$

with fish moments capped at $q_0 a_n$. Walls contribute one order of images across each of the four faces, unattenuated and without sign flip.

Receptors. Each fish carries 36 mormyromasts (10 concentrated in a $\pi/3$ chin fovea, 26 spread over the body), 24 ampullary organs and 12 knollenorgans *per conspecific*. A receptor reads the projection of the local field onto its outward normal and transduces it by a sign-preserving logarithmic compression into $[-1, 1]$,

$$\hat{s} = \text{sgn}(s) \frac{\log_{10} \text{clip}(|s|, E_{\min}, E_{\max}) - \log_{10} E_{\min}}{\log_{10} E_{\max} - \log_{10} E_{\min}}. \quad (5)$$

Mormyromasts use $[E_{\min}, E_{\max}] = [5 \times 10^{-8}, 5 \times 10^{-2}] \text{ V m}^{-1}$ and report the field of the *induced* dipoles alone: the direct discharge field is cancelled, so what remains is the electric image of objects. Images carried

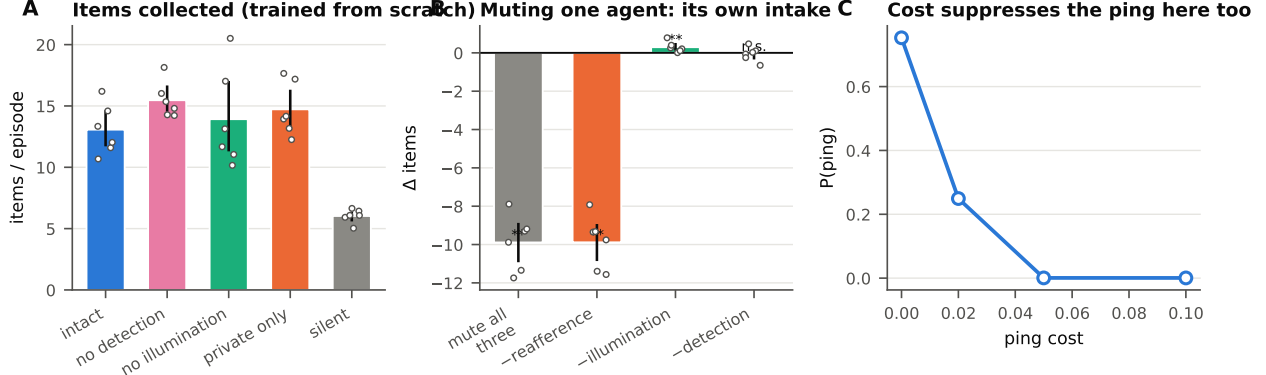


Figure 13: A minimal dual-use foraging world with no fish physics: a PING senses, illuminates neighbours and is registered as an event. The muting effect decomposes the same way as in the electric-fish environment, and cost suppresses the ping the same way.

by a conspecific’s discharge rather than one’s own are rescaled by 100, as upstream. Ampullary organs ($[2 \times 10^{-10}, 2 \times 10^{-8}] \text{ V m}^{-1}$) sense only the permanent intrinsic dipoles and are therefore independent of any discharge; a conspecific discharge degrades them (multiplicative noise fraction rising from 0.05 to 0.5), so emitting jams a neighbour’s passive sense. Knollenorgans are all-or-none: a slot fires with the sign of the local field when $|s| > 2 \times 10^{-7} \text{ V m}^{-1}$ (a detection range of $\approx 100 \text{ cm}$ at unit gain). Crucially, the knollenorgan array is indexed *by emitter*, so sender identity and bearing are preserved, and each slot carries a noisy scalar of the emitter’s relative body size.

Task and reward. Food is distributed in four Gaussian patches ($\sigma = 6 \text{ cm}$) and is not replenished. A fish eats an item within 2 cm and $\pm\pi/4$ of its heading; contested items go to the larger fish. The per-step reward is

$$r_i = 10 n_i^{\text{eat}} + \frac{d_i^{\text{prev}} - d_i^{\text{food}}}{W+H} - 5 b_i^{\text{bitten}} (1 + s_{\text{biter}} - s_i) - 0.001 b_i^{\text{bite}} - 0.5 c_i^{\text{coll}} \underbrace{- \lambda e_i}_{\text{new}} \underbrace{- \rho k_i}_{\text{new}}, \quad (6)$$

where $e_i \in \{0, 1\}$ indicates a discharge and k_i a predator strike. The first five terms are upstream’s; λ (metabolic cost) and ρ (predation) are introduced here. Under cooperative harvesting a fraction α of each neighbour’s intake is added to r_i .

Predators. A predator perceives only fish that discharged on the current step and lie within 60 cm , weights them by $1/(d+1)$, and advances 0.45 cm toward the loudest. Within 4 cm it strikes the nearest fish—not necessarily the emitter—and enters a 40-step refractory period before respawning at a random location. A silent fish is invisible, but a silent fish beside a discharging neighbour is not safe, so signalling imposes a risk externality on the group. The predator carries a strong intrinsic dipole and is therefore detectable passively at $\approx 25 \text{ cm}$, far inside the range at which it detects discharges.

K.2 The channel algebra

A discharge by fish j at time t is routed through three independent binary masks:

V_j^{self}	does j ’s pulse produce j ’s own electric image?
V_j^{illum}	does j ’s pulse produce images on <i>other</i> fish’s mormyromasts?
V_j^{det}	does j ’s pulse register on other fish’s knollenorgans?

Setting all three equal to the emitted action recovers the biological default. The conditions used here are: PRIVATE ($V^{\text{illum}} = V^{\text{det}} = 0$), CUE-ONLY ($V^{\text{det}} = 0$), SIGNAL-ONLY ($V^{\text{illum}} = 0$), SOCIAL-ONLY ($V^{\text{self}} = 0$), and MUTE (all zero, the upstream intervention). Three further manipulations act on what receivers hear without changing what the emitter does, so that any change in receiver behaviour is attributable to the public channel alone:

- **Phantom.** Receivers perceive discharges from j drawn from an exogenous Bernoulli process that j did not choose.
- **Replay.** j 's public train is replaced by the train the same policy produced in a *different arena*. Marginal rate, burst structure and inter-pulse-interval statistics are preserved exactly—these are the same trains—while the contingency between a pulse and the receiver's current world is destroyed. A context-swapped variant pairs each arena with the one whose food-proximity time series is most opposite.
- **Scrambling.** Timing scrambling circularly shifts each train, preserving rate and destroying temporal alignment; identity scrambling permutes the knollenorgan slots, preserving rate and timing and destroying sender identity.

K.3 Learning

Agents are trained with recurrent multi-agent PPO [34, 35] under centralised training with decentralised execution: one shared actor (2-layer MLP encoder, LayerNorm, tanh, 128 units $\rightarrow$ GRU with 128 units [36]) and a centralised critic that observes all agents' concatenated observations. The action space is hybrid—two Gaussians for thrust and turn, two Bernoullis for discharge and bite. Keeping the discharge decision explicitly Bernoulli matters for the analysis, because its probability is exactly the quantity we intervene on. GAE [37] with $\gamma = 0.99$, $\lambda = 0.95$; PPO clip 0.2; 4 epochs, 4 minibatches; Adam at 3×10^{-4} ; 512 parallel arenas; 64-step truncated BPTT; 18×10^6 agent-steps per run. Every condition is trained from scratch with independent seeds; *no seed selection of any kind is applied*, and all reported statistics pool over all seeds.

K.4 Measurement

Interventional causal influence. Jaques et al. [8] define the influence of agent j on agent i as the KL divergence between i 's policy conditioned on j 's actual action and its policy marginalised over j 's alternatives, and estimate it with a learned model of other agents. Owning the simulator, we evaluate the counterfactual directly:

$$\text{CIE}_t^{j \rightarrow i} = D_{\text{KL}}[\pi_i(\cdot \mid h_t^i, o_t^i[\text{do}(e_j=1)]) \parallel \pi_i(\cdot \mid h_t^i, o_t^i[\text{do}(e_j=0)])], \quad (7)$$

where $o_t^i[\cdot]$ denotes re-rendering i 's entire sensory input under the intervention with the recurrent state h_t^i and every other cause held on its factual path. The KL is analytic for the factored Gaussian $\times$ Bernoulli policy. The null is the same quantity computed between two re-renderings of the *same* intervention under independent sensory-noise draws: the divergence the policy shows when nothing about the message changed.

Positive signalling and positive listening. Positive signalling [7] is the mutual information between a window descriptor of the emitter's pulse train (discharge count over 21 steps, quantised) and the emitter's private state, normalised by the state entropy and computed *conditionally on* distance-to-nearest-neighbour decile. The stratification is essential: in a patchy world a fish near a neighbour is also near food, so an unconditioned mutual information can be manufactured entirely by geometry. Estimation is plug-in with Miller–Madow bias correction, with a permutation null. Positive listening is the L^1 divergence between the receiver's factual policy and its policy under a time-shifted message, which preserves the marginal rate and destroys contingency—the null recommended by Lowe et al. [7] in preference to a silent channel, which is out of distribution.

Message content. What a receiver could recover is decoded *only* from receiver-observable quantities: the number of pulses it heard from that emitter in a trailing window and whether it heard anything. Ground-truth state is never given to the decoder. Targets are the number of live food items within 10 cm of the emitter, whether the emitter eats within the next 41 steps, emitter body size (dominance), emitter turn command (intended movement), and whether a predator is within 25 cm (danger). We report cross-validated R^2 or AUC minus the same quantity computed on time-shifted messages, so that any score attributable to marginal rate alone is subtracted off.

Sender Shaping Index. A signal, on Scott-Phillips’s definition, requires that production be shaped by its effect on receivers. We train matched populations in a *hearing* world and a *deaf* world (knollenorgans disabled throughout training) and compare their emission policies on a common, condition-neutral batch of observations:

$$\text{SSI} = \frac{\mathbb{E}_c D_{\text{JS}}[\pi_{\text{emit}}^{\mathcal{H}}(\cdot | c) \parallel \pi_{\text{emit}}^{\mathcal{D}}(\cdot | c)]}{\mathbb{E}_c D_{\text{JS}}[\pi_{\text{emit}}^{\text{seed } a} \parallel \pi_{\text{emit}}^{\text{seed } b}]} - 1. \quad (8)$$

The denominator is the within-world seed-to-seed divergence, so $\text{SSI} \leq 0$ means the emission policy differs no more between worlds than between seeds of the same world—i.e. nobody listening changes nothing, and the pulse is a *cue*. $\text{SSI} > 0$ means production is shaped by reception.

Payoff consequences and the honesty classification. Every channel condition is evaluated against the intact channel *paired within arena*: the two rollouts share the initial food layout and the sensory noise stream, so differencing removes essentially all layout variance. Confidence intervals are 2000-sample bootstraps over arenas. Comparing the intact channel with a rate-matched replay isolates the payoff value of the signal’s *contingency*. Following Maynard Smith and Harper [4] and Dawkins and Krebs [38], a condition is labelled COMMUNICATION when contingency benefits both parties, MANIPULATION when it benefits the sender at the receiver’s expense, and EXPLOITED CUE when it benefits the receiver at the sender’s expense.

K.5 Validation against the reference implementation

Every stage of the sensory pipeline is checked against the original NumPy code in double precision on random scenes (`scripts/validate_physics.py`); results are in Table 1. Agreement is at floating-point round-off, and the knollenorgan and ampullary pathways agree bit-for-bit.

K.6 Deliberate divergences from the reference implementation

We record these explicitly because they bear on how far our results transfer.

1. **Wall-image amplitude.** Upstream’s configuration file sets a reflection coefficient of 0.95, but `ElectricScene._build_refl` never forwards it, so the code that produced the published results uses unattenuated images. We match the code, not the configuration file. Using 0.95 instead shifts the corollary-discharge baseline by $\sim 1.4 \times 10^{-6}$ relative and changes ampullary readings qualitatively, because those are a small difference of large numbers.
2. **Fixed arena and episode length.** Upstream samples arena dimensions per episode from $[40, 300]$ cm and does not report episode length. We fix a 60×60 cm arena and 512 steps so that conditions in a large factorial sweep are exactly comparable. Our conclusions are therefore about one arena scale.
3. **Collision resolution** is synchronous (all fish propose, then all resolve) rather than in agent order.
4. **Network size.** 128-unit GRU rather than 512, and 18×10^6 agent-steps per run rather than 5×10^6 environment steps, chosen to fit 354 independent training runs into the compute budget.
5. **Sensory noise** is present (multiplicative, fractions given above), following the upstream code; the upstream paper text describes a noiseless model.
6. **No seed selection.** Upstream selects a single seed (S1) by an ethological criterion and reports analyses on it. We pool over all seeds.